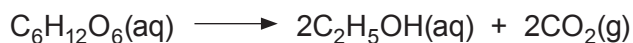


11. (a) Ethanol can be prepared by the fermentation of glucose, $C_6H_{12}O_6$. The equation for this reaction is as follows.



- (i) A student wanted to prepare ethanol and was given the following instructions.

- Mix glucose and water.
- Put the mixture into a stoppered bottle.
- Leave the bottle for several weeks.
- Separate off the ethanol.

Suggest how the student should modify these instructions in order to obtain a good yield of pure ethanol. [3]

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- (ii) Calculate the atom economy of this process as a method of preparing ethanol. [2]

Atom economy = %

- (iii) Ethanol can also be made by the hydration of ethene.

Describe **one** environmental consideration in choosing between fermentation and hydration as methods for ethanol production. [2]

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(b) Alcohols are classified as being primary, secondary or tertiary.

(i) Explain the meaning of *primary* as applied to alcohols. [1]

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(ii) Give the structural formula and name of a tertiary alcohol that contains five carbon atoms. [2]

Name

(iii) Describe **one** test that can be used to investigate whether an alcohol is secondary or tertiary. Include the reagents used and the observations for both types of alcohol. [3]

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END OF PAPER

